

Logistics & Shipment Tracking System

Final Database Architecture Document

Table: users

Column Name	Data Type	Constraints
user_id	BIGINT	PK, AUTO_INCREMENT
name	VARCHAR(100)	NOT NULL
email	VARCHAR(150)	UNIQUE, NOT NULL
password	VARCHAR(255)	NOT NULL
phone_number	VARCHAR(15)	UNIQUE, NOT NULL
role	ENUM	NOT NULL
status	ENUM	DEFAULT 'ACTIVE'
created_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP
updated_at	TIMESTAMP	AUTO UPDATE TIMESTAMP

Table: cities

Column Name	Data Type	Constraints
city_id	BIGINT	PK, AUTO_INCREMENT
city_name	VARCHAR(100)	UNIQUE, NOT NULL
value	DECIMAL(10,2)	NOT NULL

Table: pricing_rules

Column Name	Data Type	Constraints
pricing_id	BIGINT	PK, AUTO_INCREMENT
base_charge	DECIMAL(10,2)	NOT NULL
weight_rate	DECIMAL(10,2)	NOT NULL
weight_max	DECIMAL(10,2)	NOT NULL
updated_at	TIMESTAMP	AUTO UPDATE TIMESTAMP

Table: shipments

Column Name	Data Type	Constraints
shipment_id	BIGINT	PK, AUTO_INCREMENT
customer_id	BIGINT	FK → users.user_id
agent_id	BIGINT	FK → users.user_id
source_city_id	BIGINT	FK → cities.city_id

destination_city_id	BIGINT	FK → cities.city_id
pickup_address	TEXT	NOT NULL
delivery_address	TEXT	NOT NULL
package_type	VARCHAR(50)	NOT NULL
package_description	VARCHAR(255)	NULL
weight	DECIMAL(10,2)	NOT NULL
order_date	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP
expected_delivery_date	DATE	NULL
actual_delivery_date	TIMESTAMP	NULL
price	DECIMAL(10,2)	NOT NULL
status	ENUM	NOT NULL
created_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP
updated_at	TIMESTAMP	AUTO UPDATE TIMESTAMP

Table: payments

Column Name	Data Type	Constraints
payment_id	BIGINT	PK, AUTO_INCREMENT
shipment_id	BIGINT	UNIQUE, FK → shipments.shipment_id
transaction_id	VARCHAR(100)	UNIQUE, NOT NULL
gateway_name	VARCHAR(50)	NOT NULL
payment_method	VARCHAR(50)	NOT NULL
amount	DECIMAL(10,2)	NOT NULL
payment_status	ENUM	NOT NULL
payment_date	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP

Table: complaints

Column Name	Data Type	Constraints
complaint_id	BIGINT	PK, AUTO_INCREMENT
shipment_id	BIGINT	FK → shipments.shipment_id
customer_id	BIGINT	FK → users.user_id
assigned_to	BIGINT	FK → users.user_id
complaint_type	VARCHAR(100)	NOT NULL
description	TEXT	NOT NULL
status	ENUM	NOT NULL
resolution_remarks	TEXT	NULL
created_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP
updated_at	TIMESTAMP	AUTO UPDATE TIMESTAMP

resolved_at	TIMESTAMP	NULL
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Database Relationships

users.user_id → shipments.customer_id
users.user_id → shipments.agent_id
cities.city_id → shipments.source_city_id
cities.city_id → shipments.destination_city_id
shipments.shipment_id → payments.shipment_id
shipments.shipment_id → complaints.shipment_id
users.user_id → complaints.customer_id
users.user_id → complaints.assigned_to